

Week 3 Homework

Cross-Cultural Market Basket Analysis

Association rules, cultural context, and localized BI decisions

Prof. Jason J. Jung
Department of Computer Engineering, Chung-Ang University

Homework at a glance

Compare product associations across two or more observable contexts.

01

SELECT

~~Choose cultural~~
groups or
consumption contexts

02

MINE

Generate association
rules with Apriori

03

COMPARE

Evaluate support,
confidence, lift, and
sample size.

(evaluate)

04

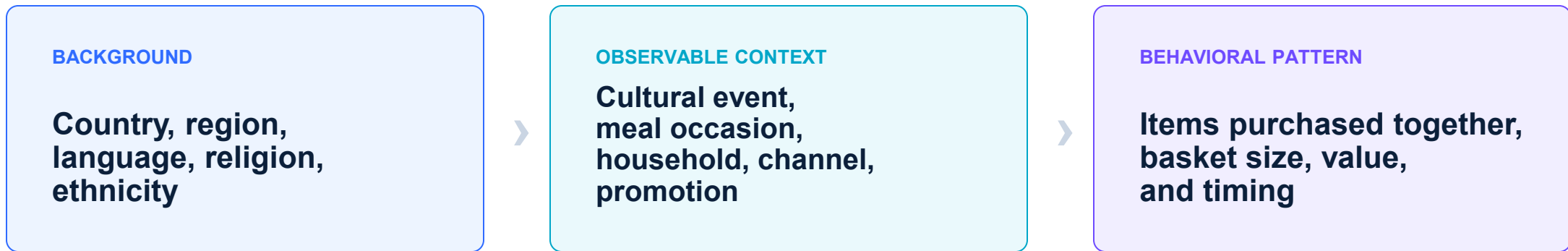
DECIDE

Recommend one
localized BI action
and one KPI.

Central principle: cultural variables support hypotheses. They do not establish causal explanations by themselves.

Cultural difference as observable context

Background variables become useful when linked to explicit settings and practices.



Interpretation example

Compare Ramadan-related transactions with baseline transactions. Avoid treating religion or nationality as a direct cause.

Synthetic market basket dataset

The instructor-provided workbook supports controlled comparison without exposing personal data.

900

TRANSACTIONS

6

MARKET CONTEXTS

22

PRODUCT INDICATORS

MARKET CONTEXTS

Seoul · Tokyo · Bengaluru · Istanbul · Paris · Dubai

CONTEXT VARIABLES

Meal occasion, social setting, cultural event, household type, region, channel, payment, promotion, weather, and price sensitivity

Use the Transactions sheet

The workbook contains one-hot product columns for association rule mining.

Research question options

Select one comparison that can be implemented with consistent filters and thresholds.

- | | | |
|----|----------------|---|
| 01 | EVENT | Do product associations differ between a cultural-event period and the baseline period? |
| 02 | SOCIAL SETTING | Does a family shopping context produce different baskets from solo shopping? |
| 03 | TIME | Do late-night baskets differ from breakfast baskets? |
| 04 | CHANNEL | Do delivery, online, and physical-store channels produce different rules? |
| 05 | PROMOTION | Does promotion exposure change the strength of a product bundle? |

Analysis workflow

Use the same preparation and rule thresholds for every comparison group.

01 Define

Research question and groups

02 Filter

Relevant transactions

03 Encode

One-hot item matrix

04 Mine

Apriori or FP-Growth

05 Compare

Support, confidence, lift

06 Recommend

BI action and KPI

Recommended Python libraries: pandas and mlxtend

Association rule measures

Each metric answers a different question about the rule $A \Rightarrow B$.

SUPPORT

How often do A and B appear together?

$$P(A \cap B)$$

CONFIDENCE

When A occurs, how often does B occur?

$$P(B | A)$$

LIFT

How much stronger is the rule than chance?

$$\text{Confidence} \div P(B)$$

Report transaction count with every comparison. A high lift from a very small group may be unstable.

Context comparison template

Compare the same rule under two explicitly defined contexts.

CONTEXT	N	SUPPORT	CONFIDENCE	LIFT
Context A	...	...	...	...
Context B	...	...	...	...
Difference		Δ	Δ	Δ

Keep the antecedent, consequent, thresholds, and preprocessing identical across the two contexts.

Interpretation and BI action

A useful conclusion separates the observed rule from its possible explanation.

INTERPRETATION CHECKS

- Observed association and difference
- Sample size and rule stability
- Alternative explanations such as price, availability, urbanization, weather, channel, or promotion
- Limits of synthetic data and causal claims

LOCALIZED BI ACTION

- Product recommendation or bundle
- Localized promotion or service design
- Inventory or channel adjustment
- Customer retention intervention

Add one KPI

Conversion, repeat purchase, retention, basket value, or intervention ROI

Submission package

Submit a reproducible analysis with enough evidence to review the conclusion.

01 Executable Python notebook

02 Research question and context definition

03 Data preparation and group sizes

04 Association rule comparison table

(Apriori Algorithm)

05 At least one comparison visualization

For example, # of rules w.r.t. confidence, ...

06 Interpretation and alternative explanations

07 One BI action and one KPI

expect which BI action be taken

08 // Limitations and data source //

The goal is to test cultural hypotheses without presenting stereotypes as facts.

[Optional reference: CultureBank](#)